"[I]t seems to me that in some respects Oppenheimer was overeducated in those fields which lie outside the scientific tradition, such as his interest in religion, in the Hindu religion in particular, which resulted in a feeling for the mystery of the Universe that surrounded him almost like a fog. He saw physics clearly, looking toward what had already been done, but at the border he tended to feel that there was much more of the mysterious and novel 'than there actually was. He was insufficiently confident of the power of the intellectual tools he already possessed and did not drive his thought to the very end because he felt instinctively that new ideas and new methods were necessary to go further than he and his students had already gone."

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1. The amount of light received at Earth is inversely proportional to the *square* of the distance to the supernova, so a factor 10 error in distance meant a factor 100 error in Baade's estimate of the total light output.

2. Antimatter gets its name from the fact that when a particle of matter meets a particle of antimatter, they annihilate each other.

3. It turns out that cosmic rays are made in many different ways. It is not yet known which way produces the most cosmic rays, but a strong possibility is the acceleration of particles to high speeds by shock waves in gas-cloud remnants of supernova explosions, long after the explosions are finished. If this is the case, then in an indirect sense Zwicky was correct.

4. The reason was explained in Box 4.2.

5. This may seem surprising to people who think of the nuclear force as far more powerful than the gravitational force. The nuclear force is, indeed, far more powerful when one has only a few atoms or atomic nuclei at one's disposal. However, when one has several solar masses' worth of atoms (10^{57} atoms) or more, then the gravitational force of all the atoms put together can become overwhelmingly more powerful than their nuclear force. This simple fact in the end guarantees, as we shall see later in this chapter, that when a massive star dies its huge gravity will overwhelm the repulsion of its atomic nuclei and will crunch them to form a black hole.

6. Superfluidity is a complete absence of viscosity (internal friction) that occurs in some fluids when they are cooled to a few degrees above absolute zero temperature—that is, cooled to about minus 270 degrees Celsius.

7. Actually, the maximum white-dwarf mass in Figure 5.5 (Wakano's calculation) is 1.2 Suns, which is slightly less than the 1.4 Suns that Chandrasekhar calculated. The difference is due to a different chemical composition: Wakano's stars were made of "cold, dead matter" (mostly iron), which has 46 percent as many electrons as nucleons (neutrons and protons). Chandrasekhar's stars were made of elements such as helium, carbon, nitrogen, and oxygen, which have 50 percent as many electrons as nucleons. In fact, most white dwarfs in our Universe are more nearly like Chandrasekhar's than like Wakano's. That is why, in this book, I consistently quote Chandrasekhar's value for the maximum mass: 1.4 Suns.